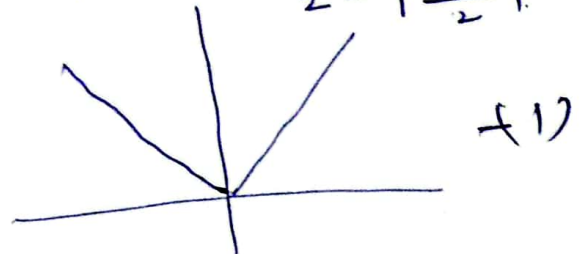


Question 1	Marks 10
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- a) Graph the function by hand, not by plotting points, but by starting with the graph of standard function involved.

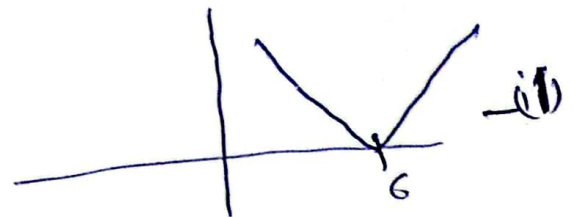
Base function:

$$y = |x|$$



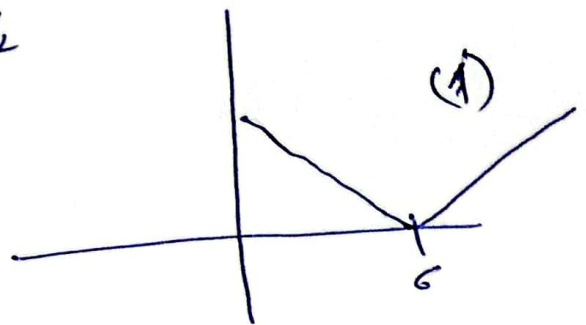
Shift rightward

$$y = |x - 6|$$



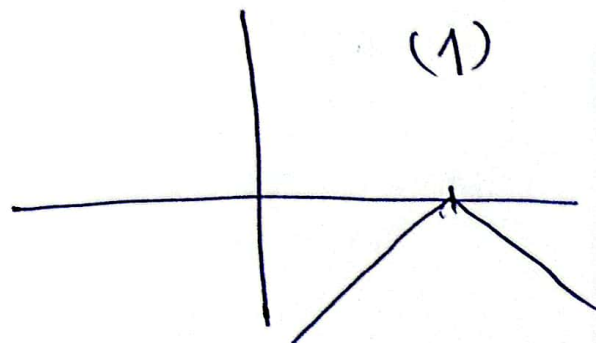
stretch horizontally by factor $\frac{1}{2}$

$$y = \left| \frac{x - 6}{2} \right|$$



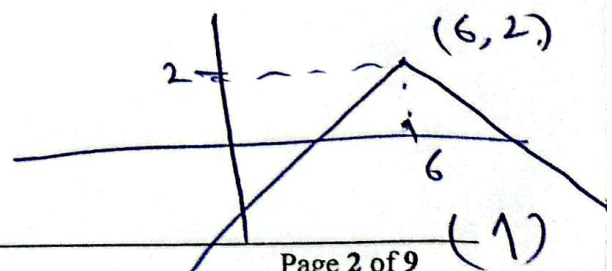
Reflection about x-axis

$$y = -\left| \frac{x - 6}{2} \right|$$



Vertical shift upward

$$y = 2 - \left| \frac{x - 6}{2} \right|$$



- b) Find equations of the function $y = \sqrt{x}$ that
- i. Shift 1 unit left and 3 units upward.
 - ii. Shift 4 units down and 2 units right.
 - iii. Reflect about the x-axis.
 - iv. Reflect about the y-axis.

(i) $y = \sqrt{x+1} + 3$ — (1, 25)

(ii) $y = \sqrt{x-2} - 4$ (1, 25)

(iii) $y = -\sqrt{x}$ (1, 25)

(iv) $y = \sqrt{-x}$ (1, 25)

Question 2

Marks 10

a) Solve the inequality $\left|3 - \frac{1}{x}\right| \leq \frac{1}{2}$

$$-\frac{1}{2} \leq 3 - \frac{1}{x} \leq \frac{1}{2}$$

$$-3 - \frac{1}{2} \leq \cancel{-3} + \cancel{3} - \frac{1}{x} \leq \frac{1}{2} - 3$$

$$-\frac{7}{2} \leq -\frac{1}{x} \leq -\frac{5}{2}$$

$$\frac{7}{2} \geq \frac{1}{x} \geq \frac{5}{2}$$

$$\frac{2}{7} \leq x \leq \frac{2}{5}$$



b) Find the solution of $|x + 3| + |2x + 1| = 6$

$$|x+3| = \begin{cases} x+3 & x > -3 \\ -(x+3) & x \leq -3 \end{cases} \quad |2x+1| = \begin{cases} 2x+1 & x > -\frac{1}{2} \\ -(2x+1) & x \leq -\frac{1}{2} \end{cases}$$

$\textcircled{1}$ $-(x+3)$ $-(2x+1)$	$\textcircled{2}$ $+(x+3)$ $-(2x+1)$	$\textcircled{3}$ $+(x+3)$ $+(2x+1)$
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$(x+3) + (2x+1) = 6$ $\textcircled{1}$

$\textcircled{1}$
 $-(x+3) - (2x+1) = 6$

$x = -10/3 \checkmark$

$\textcircled{2}$
 $(x+3) - (2x+1) = 6$

$x = 4$

$x =$
 $3x + 4 = 6$
 $3x = 6 - 4$
 $x = 2/3 \checkmark$

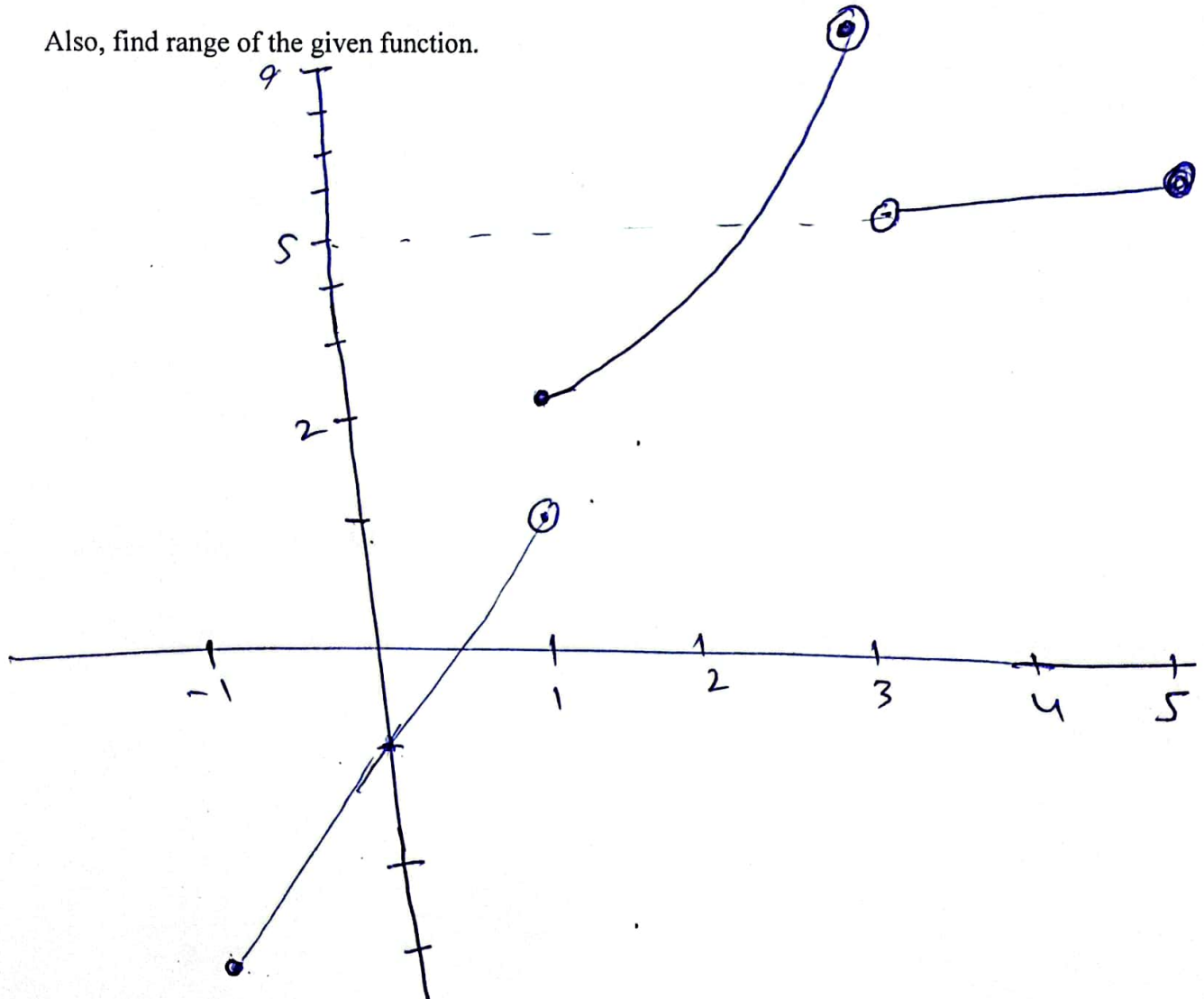
$\left\{ -10/3, 2/3 \right\}$ $\textcircled{1}$

Question 3

Marks 10

- a) Sketch the graph of the piece-wise defined function $f(x) = \begin{cases} 2x - 1 & -1 \leq x < 1 \\ x^2 + 1 & 1 \leq x < 3 \\ 5 & 3 \leq x \leq 5 \end{cases}$

Also, find range of the given function.



b) Find the domain of the function $h(x) = \frac{x+3}{4+\sqrt{x^2-9}}$

$$x^2 - 9 \geq 0$$

$$x^2 \geq 9$$

$$\pm x \geq 3$$

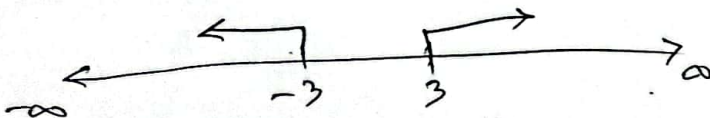
$$x \geq 3 ; -x \geq 3$$

$$x \geq 3 \quad x \leq -3$$

$$4 + \sqrt{x^2 - 9} = 0$$

$$\sqrt{x^2 - 9} = -4$$

{ }



$$(-\infty, -3] \cup (3, \infty]$$

Question 4	Marks 10
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a) Find $f \circ h$, $h \circ g$ and their domains if $f(x) = \tan x$, $g(x) = \frac{x}{x-1}$, $h(x) = \sqrt{x}$

$$f(x) = \tan x \quad \text{Domain } \mathbb{R}$$

$$g(x) = \frac{x}{x-1} \quad \text{Domain } \mathbb{R} - \{1\}$$

$$h(x) = \sqrt{x} \quad \text{Domain } [0, \infty)$$

$$\mathbb{R} \setminus \{x \mid x \neq n\pi/2, x \neq 0\}$$

$$f \circ h = \tan \sqrt{x}$$

$$h \circ g = \sqrt{\frac{x}{x-1}}$$

$$(-\infty, 0] \cup [1, \infty)$$

b) Find inverse of the function $g(x) = \frac{x+1}{x-2}$

$$y = g(x)$$

$$g^{-1}(y) = x$$

$$y = \frac{x+1}{x-2}$$

$$y(x-2) = x+1$$

$$xy - 2y = x+1$$

$$xy - x = 2y+1$$

$$x(y-1) = 2y+1$$

$$x = \frac{2y+1}{y-1}$$

$$g^{-1}(y) = \frac{2y+1}{y-1}$$

$$g^{-1}(x) = \frac{2x+1}{x-1}$$